

NeuroCore Repo Snapshot

Curated PDF for repository presentation, based on the uploaded ZIP archive. This version highlights work actually present in the notebooks and promotes only results that appear in runtime outputs or saved processing summaries.

Archive snapshot audited

55 notebooks, 17 CSV files, 8 PDFs, 10 ZIP packages (excluding macOS duplicate metadata).

Method used for this snapshot

Notebook outputs were checked against the code. Headline numbers were kept only when they appear in executed outputs. Values typed manually inside report cells were treated as illustrative and not promoted as validated results.

What this PDF is for

This is a repo-facing summary: concise, presentable, and conservative. Its goal is to show the scope of the work already done, the most credible quantitative signals currently visible in the archive, and the notebooks that deserve to remain central for the next research pass.

Verified highlights

Notebook	Verified result	Why it matters
EEGMMIDB.ipynb	Mean AUC: 0.6387 after processing the full EEGMMIDB directory. The notebook output shows the run over the EDF hierarchy, then saves <i>signature.csv</i> and <i>report_signature.pdf</i> .	This is the clearest archived classification-style result tied to a full-dataset pass. A separate report-generation cell in the same notebook contains <i>auc_final</i> = 0.8327, so that later value was not promoted.
Alzheimer_EEG.ipynb	Repeated nested CV: Logistic Regression meanAUC = 0.708 (std 0.045), pooledOOF = 0.739 ; SVM-RBF meanAUC = 0.588 (std 0.068), pooledOOF = 0.688 .	This is one of the most methodologically mature notebook outputs in the archive because it reports repeated nested CV rather than a single split.
PURSUE_LRP_ER_N_Flanker.ipynb	Best prediction horizon: 6 with AUC = 0.6155 . The output also reports state counts: STABLE 82, TRANSITION 57, CRITICAL 25.	Useful as a transition-analysis example: it links regime logic, thresholding, horizon search, and exported analysis artifacts.
ds005557.ipynb	Streaming AUC results saved across multiple EDF passes. Visible outputs range from 0.2405 to 0.8015 , with a rough mean around 0.487 across the AUC lines visible in the archived output.	Important not because the average is strong, but because it demonstrates large-scale progressive processing on a heavy dataset with saved streaming outputs.

Work that is clearly implemented, even when the final metric is not yet locked

Several notebooks contain serious pipeline structure but do not preserve a final, clean headline metric in the archived output cell stream. These still matter for the repo because they show architecture, validation intent, and next-step readiness.

Notebook	What it contributes	Repository action
neurocore_chbmit_functional_stability_v1.ipynb	Contains explicit CHB-MIT validation stages for change-point vs seizure matching, step-B tuning, and final consolidation blocks.	Keep
SeizelT2.ipynb	Large-scale seizure-oriented exploration with thousands of processed files and exploratory invariant reports. The 18.5-minute claim was not confirmed as a saved validated output.	Keep, but do not headline unvalidated lead-time claims
siena_pipeline_results.ipynb	Relevant title and likely central dataset role, but the archived notebook does not preserve a final output summary in the visible cell outputs.	Keep as supporting notebook
eegmmidb_pipeline_results.ipynb	Includes automatic reporting over 56 CSV files and produces a PDF/CSV summary on Desktop.	Keep as supporting reporting utility

Quality control notes

This archive includes both strong runtime outputs and presentation-oriented cells. For a public repo snapshot, the most important distinction is between values computed by the pipeline and values inserted manually into a later reporting cell.

Status	Interpretation
Accepted as validated snapshot material	Values printed in executed notebook outputs, repeated nested-CV summaries, saved streaming summaries, and dataset-processing logs.
Not promoted as headline results	Numbers that appear only as manually assigned variables in report-generation cells. Example: the later EEGMMIDB report cell with <i>auc_final</i> = 0.8327.
Exploratory but not yet locked	Very high values without enough visible validation context, theoretical-baseline contrasts, and unconfirmed seizure lead-time claims.

What the repo can honestly claim right now

NeuroCore already demonstrates:

- multi-dataset experimentation across EEG / IEEG / clinical and task-based data;
- transition-sensitive analysis pipelines that produce saved artifacts, thresholded state summaries, and repeatable notebook outputs;
- at least one clear whole-dataset EEG classification result (EEGMMIDB mean AUC 0.6387);
- a stronger clinical-style modeling pass in Alzheimer_EEG.ipynb with repeated nested CV (LogReg meanAUC 0.708, pooledOOF 0.739).

Recommended core set for the next cleaned repo iteration

Keep	Reason
EEGMMIDB.ipynb	Best clean repo-facing baseline result visible in archive outputs.
Alzheimer_EEG.ipynb	Most rigorous validation-style notebook currently visible in the archive.
PURSUE_LRP_ERN_Flanker.ipynb	Good transition / horizon-search showcase with exported artifacts.
ds005557.ipynb	Shows scaling behavior and progressive processing on a large dataset.
neurocore_chbmit_functional_stability_v1.ipynb	Core seizure-validation architecture for future locking of pre-ictal claims.
SeizelIT2.ipynb	Exploratory seizure-scale notebook worth retaining, with stricter claims discipline.

Repository-ready closing statement

This codebase already contains substantial work: full-dataset passes, exported reports, thresholded state analysis, repeated CV experiments, and several domain transfers. The right presentation is not maximal hype; it is disciplined evidence. Framed that way, the repo shows a real research program in progress with a visible

computational core, credible intermediate results, and a clear path to stronger validation.